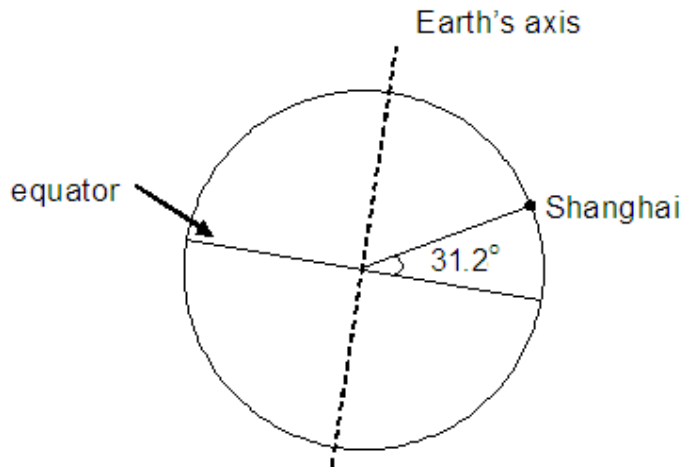


- 9 A person is located near the city of Shanghai which has a latitude of 31.28° N.

Assuming that the Earth is a sphere of radius 6380 km, find the linear velocity of the person due to the rotation of the Earth about its axis.



- A** 240 m s^{-1} **B** 337 m s^{-1} **C** 397 m s^{-1} **D** 464 m s^{-1}
- 10 The density of a sample of helium gas at the pressure of 100 kPa is 0.180 kg m^{-3} . The